Supplementary Information: An artificial general engineer that learns software and physical engineering through verified trial and error

SI 1. Parameter ranges and problem generators

All datasets are generated deterministically from uniform sampling over the ranges below (SI units; angles in degrees). The generator (`physx/dataset.py`) is committed and reproducible: the same seed yields bit-identical problems.

Domain	Parameters (range)	Answer
Projectile	$v_0 \in [5, 30]$ m/s, $\alpha \in [10°, 80°]$	range (
Pendulum	$L \in [0.5, 3]$ m, $\theta_0 \in [5°, 60°]$	period
Spring–mass	$k \in [5, 50]$ N/m, $m \in [0.2, 5]$ kg, $A \in [0.1, 1]$ m	$\omega = \sqrt{k}$
Beam	$L \in [1, 6]$ m, $P \in [10^3, 5 \times 10^4]$ N, $E \in [69, 210]$ GPa, $I \in [10^{-7}, 2 \times 10^{-5}]$ m^4, $h \in [0.05, 0.4]$ m	$w_{\max} =$
Cantilever	same five ranges as beam	$w_{\max} =$
Burgers	$\nu \in [0.02, 0.1]$, $A \in [0.6, 2]$, $\sigma \in [0.2, 0.5]$	peak $u($
RC circuit	$R \in [10^3, 10^6]$ Ω, $C \in [10^{-6}, 10^{-3}]$ F, $V_0 \in [1, 24]$ V	$\tau = RC$
Heat2D	$A \in [100, 500]$ K, $k, l \in [1, 3]$ (integer mode numbers)	peak T

Reference trajectories are sampled at $T = 50$ points from the closed-form solutions on grids fully determined by the parameters (time for the dynamical domains, position along the span for the bending domains), so the model never predicts a redundant coordinate.

SI 2. Independent numeric verifiers

Each closed form is cross-checked by a verifier implemented without reference to it, so the agent's verification gate is not a tautology.

Domain	Verifier	Grid	Tolerance
Projectile	Euler integration of ballistic ODE	$\Delta t = T/2000$	2×10^{-3}
Pendulum	RK4 integration; period by step-halving convergence	$\Delta t = 10^{-4}$	10^{-3}
Spring–mass	ODE residual $m\ddot{x} + kx$ by central differences	400 points	10^{-3}
Beam	FD solution of $EIw'''' = q$ (ghost points at ends)	400 nodes	2×10^{-2}
Cantilever	FD double integration of $M(x) = -P(L - x)$ (ghost at clamp)	400 nodes	2×10^{-2}
Burgers	finite-volume upwind for $u_t + (u^2/2)_x = \nu u_{xx}$	2,000 cells	5×10^{-2}
RC circuit	Euler integration of $\dot{V} + V/\tau = V_0/\tau$	4,000 steps	2×10^{-3}
Heat2D	sparse FD Poisson solve, 5-point stencil, Dirichlet rows	48^2 grid	2×10^{-3}

The Burgers domain is the deliberately hard case: a nonlinear conservation law whose solutions steepen into shocks as ν decreases. Its "closed form" is the Cole–Hopf transform $u = -2\nu\, \phi_x/\phi$ with ϕ the heat evolution of $\phi_0(x) = \exp(-(1/2\nu) \int_{-\infty}^{x} u_0)$, evaluated by direct kernel quadrature on a fine grid (an FFT derivative would ring on the non-periodic, many-orders-of-magnitude ϕ_0). The finite-volume verifier is written from the conservation form only and agrees with the Cole–Hopf reference to $\sim$1% on the model grid at 2,000 cells. The predicted trajectory is the flattened 25×8 (x, t) field; the physics-consistency layer evaluates the Burgers residual $u_t + uu_x - \nu u_{xx}$ pointwise on that true grid, and the scalar answer (the peak of the final field) is the shape-norm scale, with the field reconstructed as shape $\times$ A at inference (the field peaks at the initial amplitude A).

The cantilever finite-difference solver deserves note: because the clamped end carries two boundary conditions ($w(0) = 0$, $w'(0) = 0$), the second-order moment equation is discretized with a ghost point $w_{-1} = w_1$ folded into the ODE at $x = 0$ and a free-end ghost w_n, so that every interior index carries a second-difference stencil. Pinning w_0 and w_1 as separate rows instead leaves a spurious linear null mode (a kinked ramp invisible to the interior stencil), which we diagnosed during development and avoided.

SI 3. PhysFormer architecture

- **Input tokens:** one token per parameter, embedding of the parameter id plus a linear projection of the z-scored value plus a learned positional embedding (capacity for up to 16 parameters).
- **Reasoning layers:** standard transformer encoder, GELU activation, dropout 0.05, layer norm, batch-first.
- **Answer head:** two linear layers (hidden width = d_{model}), GELU in between; output in $\log_{10}$ space for beam, cantilever, and RC.
- **Trajectory head:** MLP with two hidden layers (width `traj_hidden`, 64–128), output $T \times C$ where $T = 50$ for the one-dimensional domains, $T = 25 \times 8 = 200$ for the Burgers field, and $T = 20 \times 20 = 400$ for the heat plate, and $C = 2$ for projectile/pendulum/spring, $C = 1$ otherwise.
- **Baselines:** an MLP encoder with the same capacity budget replaces the attention stack (same heads, data-only loss), and a "no-physics" transformer disables the residual term; both are trained on identical budgets (see SI 4) so attention and physics are isolated.
- **Physics-consistency layer:** computes the governing-equation residual of the de-scaled predicted trajectory on the true grid (residuals.py); the loss uses its scale-free normalized form ρ/ρ_{det}, ramped in after epoch 10. A per-sample hard-example weighting was implemented and ablated; at this data scale it destabilized training (see SI 4 note) and is not used in the released models.
- **Law-Conditioned Attention (LCA):** the multi-law generalist (`PhysFormerLCA`) replaces the per-domain parameter embedding with a shared physical-quantity vocabulary (13 tokens: length, force, modulus, inertia, thickness, speed, angle, stiffness, mass, amplitude, resistance, capacitance, voltage), so beam and cantilever present the identical token sequence. Each sample also carries its law signature (Table S1); an MLP embeds it into a law vector, and

every encoder layer cross-attends to it (residual + layer norm). The ablation feeds a constant signature so the conditioning stream exists but carries no equation information. One shared answer head and one shared trajectory head serve all six 50-step laws; trajectories are normalized per channel by their own peak and answers are z-scored per domain.

Training: AdamW (lr $= 10^{-3}$, weight decay 10^{-4}), global gradient norm clipping at 1.0, batch size 32, validation = last 1/8 of the generated problems (by generation order). The seed controls data generation, batch shuffling, and weight initialization, so multi-seed runs are proper statistical replicates; reported errors are mean ± std over five seeds. All seeds, hyperparameters, datasets, weights, and per-epoch metrics are committed.

SI 4. Training runs referenced in the paper

Run	Domain	Epochs	Samples	d_{model}	Layers
baseline matrix (Table 2)	beam / cantilever / projectile / burgers / heat2d	60	256	48	3
abl: physics on/off (Fig. 6)	beam	40	256	48	3
abl: global norm (Fig. 6)	beam	40	256	48	3
projectile (early)	projectile	30	1,500	48	3

Per-epoch metrics for every run are available as training logs in `physx/models/*.log`; the ablation and baseline figures in the main text are parsed directly from those logs (`fig6_data.json`, `fig9_data.json`). The headline models used for the generalization figures are the median-seed runs of the baseline matrix, promoted to `physx/models/<domain>.pt` by `physx/run_matrix.py`.

SI 5. Agent loop and skills

The agent (Node.js, zero dependencies) executes a plan of skill invocations inside a sandboxed working directory. Skills: `inspect` (file tree), `read` (file contents, capped), `search` (regex, line numbers), `edit` (create/rewrite/append, path-confined), `run` (shell with 120 s timeout and denylist), `verify` (gated check), `physx` (spawns the solver without a shell, JSON via argv). Steps annotated `expect: ok` gate the loop: a failed gated step fails the mission and writes a lesson to the episodic journal.

The verification-gate benchmark (Table 3, `bench/gate_bench.js`) runs the same missions twice: with the gate enforced (`enforceGate: true`) and with it disabled (`false`), in which case gated failures are not distinguished from successes and the agent trusts its own claims. Ground truth is computed independently of the agent: the test suite is run by the benchmark itself, and physics missions are checked against the solver's independent numeric verification.

Safety: path containment on every file operation; command denylist (`rm -rf /`, `git push`, `git reset --hard`, privilege escalation, disk formatting, shell-pipe-to-sh) unless `AGE_ALLOW_DANGEROUS=1`; python bytecode caches purged before every python command so the verification gate always recompiles from source (a same-length, same-second edit can otherwise be masked by a stale `.pyc`, which we observed in practice).

SI 6. Figure data

The exact numbers used in Figures 3–11 are stored as JSON alongside the figure scripts (`paper/fig/fig*_data.j` and every ablation and baseline curve is parsed from real training logs, so all quantitative claims in the manuscript are traceable to committed artifacts. The significance tests are reproduced from the training logs by `physx/significance.py` (output `paper/fig/significance.json`), and the DeepXDE comparison by `physx/deepxde_baseline.py` (output `paper/fig/deepxde_comparison.json`). The gate benchmark results are in `bench/gate_bench_results.json`.

SI 7. External PINN baseline (DeepXDE)

DeepXDE 1.15.0 (PyTorch backend, vendored in `vendor/`) was trained per instance on three representative Burgers problems (canonical ν=0.05, A=1.5, σ=0.3; mild ν=0.10, A=0.8, σ=0.45; shock ν=0.02, A=1.8, σ=0.25), one fresh 2–50×4–1 tanh FNN per problem, 2,000 collocation points, Adam 12k steps then L-BFGS 2k steps, exact Cole–Hopf boundary data at $x = \pm 1$, identical for every instance. Both models are evaluated with identical metrics on the model grid: final-time peak error, maximum pointwise error over the full (x, t) solution relative to the final-time peak, and the same finite-difference governing-equation residual that trains PhysFormer's physics layer.

	Peak error (%)		Field error (%)		Residual	
Instance	DDE	PhysF.	DDE	PhysF.	DDE	PhysF.
Canonical	0.00	0.54	0.03	32.9	3.3×10^{-3}	2.4×10^{-4}
Mild	0.01	0.02	0.06	50.5	1.3×10^{-3}	1.7×10^{-3}
Shock	0.00	3.73	0.03	37.9	5.1×10^{-3}	1.6×10^{-3}

The per-instance PINN trains ∼19–21 minutes per problem; PhysFormer is one network trained once on 256 problems (∼1–2 minutes), then a millisecond forward pass for any instance. The generalist's residual is comparable to or lower than the converged specialist's, i.e. its predictions are as physical but less accurate. Full per-instance results are in `paper/fig/deepxde_comparison.json`, and the predicted fields in `paper/fig/deepxde_fields/`.

SI 9. Law-Conditioned Attention: vocabulary, signatures, and the multi-law experiment

The multi-law experiment trains one `PhysFormerLCA` on six of these laws (beam, cantilever, projectile, pendulum, spring, RC — all 50-step geometries) with a single shared answer head and trajectory head. Per seed, each law contributes 96 training samples (576 total) at 250 epochs; the conditioning condition uses the true signatures, the ablation uses a constant signature, everything else identical. Held-out evaluation (last 1/8 per law, per-seed) reports answer relative error, curve error, and the physics residual. Specialists are the single-law models trained at 256 samples / 60 epochs.

Results (six seeds, 36 paired runs). Curve error: LCA 0.093 vs. ablation 0.118 (21% median reduction, paired Wilcoxon $p = 0.0003$, Cliff's $\delta = +0.33$). Answer error: pooled $p = 0.41$;

Table 1: The eight governing equations, tokenized into the fixed 22-operator vocabulary. The binary signature is the LCA input for each law.

Domain	Governing equation	Operator signature
Projectile	$y = x\tan\alpha - gx^2/(2v_0^2\cos^2\alpha)$	algebraic_closed_form, gravity_coupling, quadratic_in_space, trig_of_para
Pendulum	$\frac{1}{2}L^2\omega^2 + gL(1-\cos\theta) = \text{const}$	energy_conservation, nonlinear_trig, gravity_coupling, angular_velocity_s
Spring	$m\ddot{x} + kx = 0$	second_time_deriv, linear_restoring, harmonic
Beam	$EI\,w'' = M(x)$, $M = \frac{Px}{2}$ $(x \le \frac{L}{2})$	second_space_deriv, moment_curvature, piecewise_load, linear_in_space
Cantilever	$EI\,w'' = P(L-x)$	second_space_deriv, moment_curvature, linear_in_space
RC	$\dot{V} + V/\tau = V_0/\tau$	first_time_deriv, exponential_relaxation
Burgers	$u_t + uu_x = \nu u_{xx}$	first_time_deriv, first_space_deriv, second_space_deriv, self_advection, dif
Heat2D	$\nabla^2 u = f(x,y)$	second_space_deriv, laplacian_2d, separable_modes, poisson_source

per domain, LCA wins beam (0.340 vs 0.404), cantilever (0.283 vs 0.346), projectile (0.110 vs 0.112, 5/6 seeds) and pendulum (0.021 vs 0.023), is neutral on spring (0.063 vs 0.058), and is significantly worse on RC (0.611 vs 0.509, $p = 0.031$), whose normalized trajectory is the same curve for every time constant, so its answer requires trajectory-independent parameter arithmetic on a log scale spanning six decades — reported honestly as a limitation of the shared-head generalist in this regime. Generalist-to-specialist answer-error ratios: beam 3.3×, cantilever 2.8×, projectile 1.2×, pendulum 2.6×, spring 1.1×, RC 1.7×. All per-seed values, medians, p-values, and deltas are in `paper/fig/multi_law_data.json`; model weights and logs are committed under `physx/models/lca_{real,dummy}_s*.pt`.

SI 10. Causal steering, few-shot adaptation, and physics vs. data-only

Causal steering (law swap). Beam and cantilever present identical quantity tokens and identical sampling ranges, so a held-out beam problem fed with the cantilever signature (and vice versa) differs from its native presentation only in the injected law vector. Per sample, the steering index SI $= (d_{\text{src}} - d_{\text{tgt}})/(d_{\text{src}} + d_{\text{tgt}})$ and the disruption $d_{\text{src}} - d_{\text{native}}$ use normalized trajectory errors against the exact solutions of both laws. Pooled over the n=72 per-condition samples (12 held-out problems × 6 seeds) of the beam→cantilever direction:

	LCA (real signature)	Constant-signature ablation
Steering index (median)	+0.090	−0.132
paired Wilcoxon p	< 0.0001	
Signature-induced disruption (median)	0.371	0.000
paired Wilcoxon p	< 0.0001	
Per-seed median SI (6 seeds)	−0.413	−0.584 ($p = 0.031$)

The ablation's disruption is exactly zero: with a constant signature the swap cannot change the computation, confirming the control is perfect. The LCA disruption is large and its swapped prediction lands on the target-law side of the decision boundary, but full re-targeting is not achieved (the swapped prediction's error against the target law remains ∼0.36), so steering is reported as partial. The reverse (cantilever→beam) direction is excluded from the pooled test because beam deflections are the smaller family by 16×, which makes the normalized error scale trap the index; it is reported per seed in `paper/fig/law_swap_data.json`.

Few-shot adaptation. The cantilever is held out of a five-law pretraining (96 samples per law, 120 epochs); the pretrained generalist is fine-tuned on 24 cantilever samples (25% of the 96-sample specialist budget) for 40 epochs; the from-scratch specialist receives the same 24 samples and 40 epochs. Evaluation is on the same 12 held-out cantilever problems for all conditions. Held-out answer / curve error by seed:

Seed	LCA generalist	Dummy-signature ablation	Specialist
0	0.197 / 0.137	0.239 / 0.131	0.279 / 0.273
1	0.287 / 0.134	0.437 / 0.145	0.580 / 0.284
2	0.182 / 0.172	0.153 / 0.129	0.691 / 0.282
Median	**0.197** / 0.137	0.239 / 0.131	0.580 / 0.282

Physics vs. data-only (heat2d). Identical networks, 240 epochs, trajectory weight 3, three seeds; the only difference is w_{phys} (0.05 vs. 0). Held-out field max error (per seed, then mean):

	64 samples (25%)	256 samples (100%)
Physics (w=0.05), field max err	0.512 / 0.496 / 0.523	0.316 / 0.270 / 0.246
Data-only (w=0), field max err	0.116 / 0.207 / 0.150	0.032 / 0.033 / 0.032
Physics, mean residual	0.39 / 0.37 / 0.59	0.106 / 0.079 / 0.061
Data-only, mean residual	2.60 / 4.46 / 4.04	0.815 / 0.458 / 0.347

The physics term reduces the governing-equation residual 4–8× but does not improve held-out field fidelity at this capacity; the canonical (2,3)-mode case is the easiest member of the validation distribution (5.9% field error vs. ∼29% mean).

SI 8. Statistical analysis

Significance is assessed with the two-sided Wilcoxon signed-rank test on paired runs (same domain, same seed, kinds compared within the pair). Per-domain comparisons have $n = 5$ pairs, for which the minimum achievable p under the exhaustive sign-permutation null is $2/2^5 = 0.0625$; we therefore report both per-domain p-values and the pooled test over all five domains ($n = 25$ pairs per comparison), which reaches $p < 0.001$. Effect sizes are paired Cliff's δ (the probability of one kind beating the other minus the reverse). The physics term's effect on the trajectory residual is $p < 0.001$ with $\delta = -0.92$ (large, in the physics-on direction) and a ∼19-fold median reduction; no comparison on the scalar answer reaches significance ($p \geq 0.65$, $|\delta| \leq 0.12$). All p-values, deltas, medians, and per-seed values are in `paper/fig/significance.json`.